

Game AI Final Assignment Proposal

Group 29

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1 The project

For this project we want to combine Procedural Content Generations (PCG) with AI controlled agents in the game PacMan. The idea is that the game will be adapted to the players level over different playthroughs. So if the player plays poorly the next generated game should be easier to play and vice versa. When talking about generating different PacMan games procedurally, there are a number of variables that can be adjusted for a different game. For example: the layout of the map itself, the starting position of the player, the starting position of the ghosts and possibly extra features not already present in the original game. Lastly, we also want to make the agent controlling the ghosts a variable which can be adjusted. So, when the player plays well the ghosts should be controlled by a more sophisticated agent acting more aggressively to eliminate the player.

By making some variables easier and other more difficult, we can create a wide variety of different games using procedural content generation. The eventual goal is then to create a wide variety of games, to keep it interesting to the player, as well as having the difficulty be automatically adjusted to the players level.

2 Related topics

Learning and Optimization: We aim to approach reinforcement learning in games in a different angle than usual. Many agents are trained to beat a game, but agents trained to face off against the player are less common.

Experience Modeling: Difficulty in games is approached in many different ways. In our project, we want to hit a sweet spot where the player has neither too easy of a time or far too great of a struggle.

Procedural Content Generation: Procedural content generation not only keeps each experience fresh for players, but it is also more efficient in terms of storage space. It is an enticing choice for many game developers.

Believable Behavior: Believability is a serious concern, especially with reinforcement learning and procedural content generation. We will make sure that the program creates levels that are possible to complete and not have an unnatural look.

3 Milestones

We believe there are three noteworthy milestones to aim for during this project.

- Procedurally generating the stage. We must first actually be able to generate playable, feasible stages to move forward with the rest of our project.
- Agents for use by the ghosts. They must be driven by reinforcement learning to learn and improve their behavior. We will likely have to train them against previously implemented agents for the player to set benchmarks at the beginning.
- Adaptive difficulty. Bringing the previous two points together, we will consider the agent intelligence level and procedurally generated map complexity for overall difficulty. Using metrics such as time taken to complete a stage, we will reward the difficulty agent positively or negatively in order to keep difficulty in the zone we want.

4 Technical aspects

One of the most important components we need as an API which we can use for the Procedural Generation of the PacMan game. There are many different frameworks, which are possibly suitable for our objective. However, we still need to do further research to pick one. The ms PacMan AI Framework might work well, but this does not have a built in maze generator functionality. Another option is to use the WaveFunctionCollapse repository on Github which you can use to generate maps. Only then we would need to implement the rules of PacMan ourselves.

Furthermore we need a way to measure player performance and subsequently a way to model the difficulty of the game.

Lastly we need to be able to create different difficulty levels for the agents controlling the ghosts. This could be done by creating different agents like: random, heuristic or MCTS. Alternatively, this could simply be done by tweaking a parameter within a single agent type, such as tweaking epsilon in an ϵ -greedy implementation.